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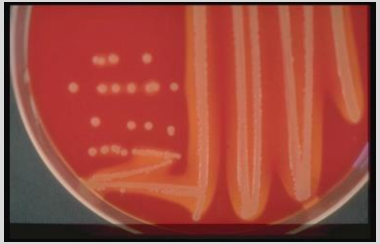
- Session Objectives:
    1. Describe the roles of microbial virulence factors and host defense mechanisms play in the prevention of or development of skin and mucous membrane infections.
    2. Describe common medically important pathogens that cause skin and mucous membrane infection.
    3. Diagnose pathogens causing skin and mucous membrane infections.
- Connections
    - ScholarRx: Bacterial Infections of the Skin

Source: Course Sy

Staphylococcus aureus



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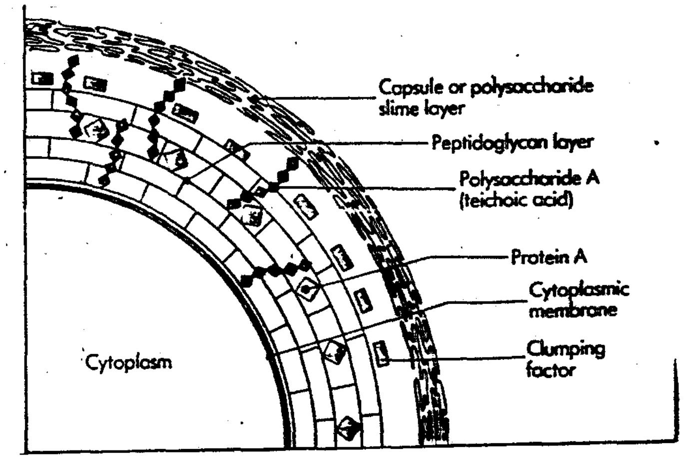
- **Coagulase positive**, beta hemolytic on blood agar
- Key virulence factors
  - **Coagulase** converts fibrinogen to fibrin, clots and a fibrin shield inhibiting phagocytosis
  - **Capsule** inhibits opsonization and phagocytosis, protects from C'-medidated leukocyte destruction
  - **Protein A** binds IgG1, IgG2, IgG4 Fc receptors, inhibits opsonization, phagocytosis, leukocyte chemoattractant, anticomplement
  - **Catalase** prevents oxidative killing by immune cells
  - Hyaluronidase, fibrinolysin, lipase, nuclease
  - **Penicillinase** opens beta lactam ring inactivating penicillin
  - Cytolytic & exfoliative toxins, **TSST**, enterotoxin



Staphylococcal cell wall structure.

Staphylococcus aureus: Other Diseases

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- Bloodstream infection
- Sepsis, septic shock
- Acute endocarditis
- Pneumomia
- Empyema
- Osteomyelitis
- Septic arthritis

- Toxic shock syndrome
  - Superantigen stimulates T-cell proliferation, cytokine production, endothelial destruction, capillary leak, shock, multiorgan failure
  - Strawberry tongue
  - Diffuse, erythematous rash

Exudative pharyngitis (usually GAS)

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- Sore throat, fever
- Diagnosis: rapid antigen, culture if negative, anti-streptolycin-O antibodies (ASO)
- Treatment: penicillin (1st line), amoxicillin, cephalexin, (anaphylaxis to PCN: clindamycin, clarithromycin)

Streptococcal Skin Diseases

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- Impetigo
  - Similar to staphylococcal impetigo, but bullae are tense and have gold crusts
- Scarlet fever
  - High fever, sore throat, sandpaper rash
  - Erythrogenic toxin gene encoded by lysogenic bacteriophage)
- Erysipelas
  - Acute, rapidly progressing, superficial skin infection
  - Bright red, warm, tender, sharply demarcated plaque with raised borders
- Cellulitis
  - Rapidly spreading, painful, erythematous skin and subcutaneous tissue infection
  - May result in bloodstream infection, endocarditis
- Necrotizing fasciitis
  - Rapidly advancing erythematous, warm bullae, crepitus (gas in tissues) and dusky skin discoloration
  - Intensely painful skin and soft tissue infection associated with fever, tachycardia, sepsis
  - May result in septic shock
  - Requires emergent surgical debridement
- Toxic shock syndrome
  - Similar to staphylococcal toxic shock

Streptococci: Non-skin MM diseases

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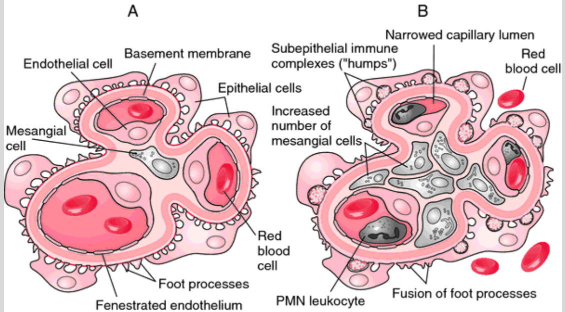
- Pneumonia (S. pneumoniae)
- Meningitis
- Post-streptococcal sequelae (GAS)
  - Rheumatic fever (GAS pharyngitis or scarlet fever)
  - Post-streptococcal acute glomerulonephritis (GAS pharyngitis)



Pseudomonas aeruginosa

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- Gram-negative, motile, oxidase-positive, non-lactose fermenting bacillus
- Colorless colonies on MacConkey agar
- Ubiquitous in moist environments
- Cell wall lipopolysaccharide (LPS) acts as an endotoxin resulting in sepsis
- Exotoxin A and esoenzyme S inhibit protein synthesis
- Infections of burns, surgical wounds, cystic fibrosis, ecthyma gangrenosum in neutropenia